Statistical Analysis Plan (SAP)

Study Title: Ice lolly in Postanesthesia Care (Icesthesia)

Study Registration Number: DRKS00037179 (2025-06-11)

Sponsor: LMU University Hospital, Department of Anaesthesiology, Marchioninistraße 15, 81377 Munich

Intervention: Ice Lolly in the postanesthesia care unit

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1. Objectives

After undergoing anesthesia, many patients experience side effects such as nausea, vomiting, sore throat, dizziness, and disorientation. These symptoms can delay recovery and prolong the time needed before patients can safely return to their hospital ward or be discharged. To manage these effects and monitor for complications, patients are transferred to a postoperative recovery unit (also known as the post-anesthesia care unit, PACU). In this setting, healthcare staff provide close observation and supportive care until patients regain stable vital signs and are sufficiently alert. Effective management in the recovery unit is essential to ensure patient safety, comfort, and timely progression through the surgical care pathway. Ice lollies may have a beneficial effect on recovery, the cold might stimulate patients, holding an ice lolly requires attention and keeps patients busy, ice lollies are often associated with positive childhood memories that can take away the focus from boredom, pain and nausea. Evidence from small or retrospective trial showed a positive effect on consumption of pain medication, however, there is no robust evidence for this intervention regarding faster discharge from the recovery room. The aim of this study is to investigate the effect of a non-pharmacological intervention in the recovery room on patients' length of stay. The hypothesis is that offering ice pops to postoperative patients leads to a faster recovery from the side effects of anesthesia, thereby reducing the length of stay in the recovery room.

1.1 Primary Objective

To assess whether application of ice lollies reduces duration of stay (in minutes) in the recovery unit postoperatively.

1.2 Secondary Objectives

To evaluate the effect of ice lolly application on:
Dosage of pain medications (opioids, NSAIDs, metamizol, paracetamol)
Pain
Incidence of nausea and vomiting
Delirium severity score (categorized as 0, 1, or ≥2) with 0 being alert and awake (normal state of consciousness)

2. Study Design

Type: Two-arm, parallel-group randomized controlled trial
Randomization: cluster randomization 1:1 allocation (days with ice lollies vs. days without ice lollies)
Sample Size: 3,140 patients (~1,570 per group)
Sample size estimation is based on the primary endpoint (PACU length of stay in minutes), the assumed minimal relevant effect size, and the planned mixed-effects modelling approach accounting for cluster randomisation.
As no prior interventional data are available, this exploratory study defined a 5% reduction in PACU length of stay as the minimal organisationally relevant effect. Assuming a mean PACU stay of 90 minutes with a standard deviation of 40 minutes, this corresponds to an

absolute reduction of 4.5 minutes (standardised mean difference Cohen's d ≈ 0.11). From an operational perspective, given a recovery unit with 25 beds and a 3:1 patient-to-staff ratio, even small reductions in average length of stay may translate into meaningful cumulative gains in daily staffing time and patient flow.
Under individual randomisation, detection of this effect using a two-sided test with $\alpha = 0.05$ and 80% power would require approximately 2,600 patients.
As the study uses cluster randomisation by day to avoid contamination, the sample size was inflated to account for within-cluster correlation using the design effect defined as:

$$\text{Design effect} = 1 + (m - 1) \times \text{ICC},$$

where m represents the average cluster size and ICC the intracluster correlation coefficient. Assuming an average of 25 eligible patients per study day (cluster size m = 25) and an ICC of 0.01, the design effect equals 1.24. The assumed ICC of 0.01 was based on conservative methodological considerations in the absence of prior empirical data. Applying this inflation factor results in a required total sample size of approximately 3,140 patients.
Recruitment will stop once 3,140 patients with complete datasets have been enrolled. Because randomisation occurs at the level of the day, recruitment cannot be stopped mid-cluster. Therefore, enrolment will continue until completion of the study day during which the required sample size of 3,140 patients is reached. As a consequence, the final sample size may slightly exceed the calculated minimum. During the planned six-month study period, approximately 5,000 patients are expected to be treated in the PACU, ensuring feasible recruitment while accounting for exclusion criteria, seasonal reductions in surgical capacity, and limited study personnel availability.
The sample size calculation was performed prior to database lock and before any outcome data were inspected.
Intervention: Postoperative patients in one large recovery room are offered an ice lolly by the nursing staff on days randomized to ice lollies plus standard of care. On days randomized to "no ice lollies", patients receive standard of care.
Randomization: An online randomization tool (studyrandomizer.com) is used to perform daily randomization. Days are randomized in a 1:1 ratio using permuted blocks with variable block sizes ranging from 2 to 6. The study personnel conduct randomization each morning at 9 am.
Blinding: Due to the nature of the intervention, blinding of clinical staff to allocation is not feasible. Nursing staff and treating physicians were not informed about the true primary outcome (PACU length of stay). Delirium score was communicated as the primary study outcome in order to minimise performance bias. The study investigator collecting outcome data was blinded to the study hypothesis related to PACU length of stay.

3. Analysis Populations

Intention-to-Treat (ITT): All randomized patients treated on randomized study days, analyzed according to the allocation of the respective study day.
Per-Protocol (PP):

- Intervention days: eligible patients who consumed the offered ice lolly.
- Control days: eligible patients treated according to standard of care without ice lollies.
Patients with major protocol deviations affecting intervention exposure will be excluded from the PP analysis.

4. Statistical Methods

Anonymized patient data is stored on a departmental server. After patient enrollment is finished, data will be quality checked. Statistical analyses will be reported using summary tables and figures. Continuous variables will be summarized by means and confidence intervals if approximately normally distributed or by medians with first and third quartiles if the distribution deviates from normality. Categorical variables will be summarized by counts and by percentages of patients. No formal statistical testing of baseline differences between groups will be performed.
All tests will be two-sided, with $\alpha = 0.05$. No adjustment for multiplicity will be applied for secondary or subgroup analyses; these analyses are considered exploratory.
Normality will be assessed visually.
The percentage of missing data is expected to be very low. If missing data remain minimal (<5%), a complete-case analysis will be performed. If missing data exceed this threshold, sensitivity analyses (e.g., multiple imputation) will be considered.
Because the study uses cluster randomization at the level of the day, all inferential analyses will account for clustering by using mixed-effects regression models including study day as random intercept.

Unless otherwise specified, all models will include treatment group and the prespecified covariates listed in Sections 5 and 6 as fixed effects. The primary analysis will follow the intention-to-treat (ITT) principle including all randomized patients according to study day allocation.

For continuous outcomes, linear mixed-effects regression models will be applied. If substantial deviation from normality is observed, an appropriate generalized linear mixed-effects model (e.g., gamma distribution with log link) will be used.

For binary outcomes, generalized linear mixed-effects models with logit link will be applied.

For ordinal outcomes, cumulative link mixed models (ordered logit models) will be applied.

For medication dosage outcomes with a high proportion of zero values, a two-part mixed-effects modelling approach will be used: first, a mixed-effects logistic regression model will analyse the probability of receiving any medication (yes/no); second, among patients receiving medication, dosage will be analysed using an appropriate mixed-effects model depending on distributional characteristics.

A per-protocol analysis will be conducted using the same modelling approach restricted to the per-protocol population.

Effect estimates will be reported with 95% confidence intervals. The choice of distributional family and link function will be guided by the nature and distributional characteristics of

the respective outcome variable. Model assumptions and fit will be assessed using graphical and statistical diagnostics.

5. Primary Outcome

Outcome: Duration of stay (in minutes), treated as a continuous variable. Distributional assumptions will be assessed as described above.

The treatment effect from the mixed-effects model will be reported as the adjusted mean difference in PACU length of stay with 95% confidence intervals and corresponding two-sided p-value. The intracluster correlation coefficient (ICC) for the primary outcome will be estimated from the fitted mixed-effects model and reported descriptively.

- The following baseline covariates will be adjusted for in the primary and secondary analyses:
  - Age group
  - Sex
  - ASA classification
  - Surgery type (neurosurgery, orthopedic surgery, ENT, plastic surgery)
  - Duration of anesthesia (in minutes)
  - Type of anesthesia

6. Secondary Outcomes

6.1 Pain and Pain Medication Dosage

- Analyzed separately for each type (opioids, NSAIDs, metamizol, paracetamol)
- Additional covariate: duration of stay in the recovery room (included as fixed effect in the mixed-effects model to account for differences in observation time within the PACU)
- NRS (Numeric rating scale) at admission and discharge (Quasi-Continuous)

6.2 Nausea and Vomiting (Binary)

- Additional covariate: duration of stay in the recovery room (included as fixed effect in the mixed-effects model to account for differences in observation time within the PACU)
- Antiemetic medication administered

6.3 Delirium Score (Ordinal: 0, 1, ≥2)

7. Subgroup and Sensitivity Analyses

- Prespecified subgroup analyses will be conducted based on:
  - Age (<12 years and >75 years)
  - Sex
  - ASA classification
  - Surgery type (neurosurgery, orthopedic surgery, ENT, plastic surgery)
  - Type of anesthesia

Subgroup analyses will be performed by including interaction terms between treatment group and subgroup variables in the respective mixed-effects models. These analyses are exploratory.

8. Software

All statistical analyses will be conducted in R (≥ 4.5) and appropriate add-on packages. Excel will be used only for data entry and management.

9. Interim analysis and safety outcomes

As the intervention is non-pharmacological and considered low risk, no specific safety monitoring procedures are planned beyond routine clinical care. There is no interim analysis planned.